SOME CONSIDERATIONS FOR SCOPE 2 EMISSIONS ACCOUNTING

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DATE: December 22, 2025
RE: Some Considerations for Scope 2 Emissions Accounting

I. SUMMARY

The Scope 2 framework has played a critical role in establishing traceability between electricity consumption and emissions outcomes within corporate reporting. While imperfect and subject to well-documented limitations, these shortcomings have often been regarded as a ”necessary evil” in advancing emissions accountability. However, it is equally important to avoid letting the pursuit of perfection undermine a framework that is broadly functional, and to weigh idealized proposals against their practical repercussions. In today's highly uncertain environment, the future viability of the Scope 2 framework hinges on reconciling its accounting conventions with the physical realities of electricity grid operations—a point articulated clearly in Dr. William Hogan's recent position paper (1). This memo advances the position that emissions accounting should align with the temporal and spatial resolution of *already* functioning power markets, thereby reflecting both technical feasibility and operational transparency.

II. CONTEXT AND CHALLENGE

Achieving hyper-resolution (e.g., asset level) in emissions accounting is fundamentally infeasible when anchored to individual load-generation matching, given the locational marginal pricing (LMP) formation process and underlying grid physics, which includes the power flow itself but also how power flow physics is represented in decision support tools (e.g., modeling assumptions that lead to understandably lower, but practically acceptable solution accuracy). This observation has significant implications for the evolving Scope 2 framework, particularly as power system operations undergo transformational changes, on both generation and load sides.

III. SOME CONSIDERATIONS FOR FUTURE SCOPE 2 FRAMEWORK

Grid Expansion Makes Hyper-Resolution, Individual Matching Infeasible

Attributional matching at the level of individual loads and generation faces increasing technical obstacles as boundaries between production and consumption in modern power systems increasingly blur. For example, battery storage resources can function as producers, consumers, or in the case of large, multi-modular battery units simultaneously as both, making discrete load/generation categorization virtually impossible. The growing deployment and aggregation of behind-the-meter resources, residential and utility-scale, and controllable high-wattage loads (e.g., heavy-duty EVs) further obscure traditional distinctions between generation and load, undermining the feasibility of granular matching. An additional issue is related to the recent move to allow co-location of large loads and power plants (2).

Market-Aligned Accounting as the Optimal Available Solution

In the evolving power sector landscape described above, aligning emissions accounting with the temporal and spatial resolution of functioning power systems and, in some cases, power markets represents the most practical approach, second only to the politically challenging option of internalizing carbon costs directly into LMP calculations. Emissions accounting aligned with *already* existing power system and market operations across temporal and spatial resolutions has the strong advantage of being both technically achievable (data and dispatch outcomes are readily available) and independently verifiable, supporting transparency and integrity. This approach also enables clear *causative linkages* between dispatch actions and emissions impacts, facilitating out-of-market settlement processes and pricing mechanisms (e.g., as relevant for PPA agreements), as well as providing interpretable insights on what could be done to reinforce ”positive” outcomes and discourage ”negative” outcomes.

Aligning emissions accounting with LMP resolution also inherently addresses deliverability constraints, i.e., it takes into account transmission limits subject to the accuracy of power flow modeling which may improve over time if more refined physics-based models are adopted. The resulting dispatch and emissions outcomes capture the effects of transmission limitations, which is a critical factor that if overlooked can introduce accounting inaccuracies and is also responsible for driving discrepancies between annual or hourly matching approaches (3). The important nuance of this alignment is that it does not have to be constrained to the current LMP resolution and can potentially be enhanced if LMPs or LMP-based settlement schemes change (e.g., potential introduction of distribution LMPs (or value-adding alternatives) or different settlement arrangements in power markets).

Data Availability and Quality

Marginal (short-run) emissions data is currently available from system operators, such as PJM, and third-party providers, enabling independent verification. While imperfections exist in current short-run marginal data sets, these map physical dispatch reality, reflect LMPs (and are thus relatable to its energy, congestion and loss components), are addressable, and do not constitute a barrier to implementation.

The current Scope 2 framework insufficiently differentiates between applications of short-run and long-run marginal emissions rates, each serving distinct purposes. Short-run marginal rates, derived from factual operational observations, are well-suited for evaluating operational decisions and enabling consequential outcome accounting. Any inaccuracies are fundamentally correctable through ex post adjustments. Long-run marginal rates are intended for guiding planning decisions and investment in new electricity procurement resources. However, as predictive instruments, they are inherently subject to forecast errors and modeling uncertainties. Thus, if short-run rates reflect physical reality of power system operations, long-run rates reflect a predicted future which may or may not materialize; a notable risk here is that when the prediction error on long-run rates is known, the investment decisions are already made and there is limited opportunity for correction. A critical gap exists in the current body of knowledge regarding the relationship between short-run and long-run rates, warranting further development to provide intuitive guidance for stakeholders.

An important delineation between the short and long-run rates is to understand the missing link of how they contribute to relating operational (short-run) and planning (long-run) carbon emissions impacts. If the underlying premise of the Greenhouse Gas Protocol is to evaluate overall carbon emissions reductions, there still needs to be a systematic framework for reconciling predicted long-run impacts with realized short-run outcomes. Thus, one critical area of concern is how both the short and long-run rates relate to the total emissions reductions goals.

IV. CONCLUDING THOUGHTS

Greenhouse Gas Protocol is not set in stone and it should adapt to align emissions accounting conventions with power system operational and market realities, specifically:

1. Abandon efforts to match individual loads with individual generation, recognizing this approach as physically infeasible in meshed networks[1]. The underlying cause of this infeasibility is the fact that power flows are driven by Kirchhoff's laws not by electricity transactions.
2. Adopt accounting methods aligned with the temporal and spatial resolution of power system operations and LMP-based markets, leveraging existing marginal emissions rate data. Greenhouse Gas Protocol could also aim to follow the gradually improving fidelity/resolution of power system operations and LMP-based markets (as well as the representation of physical models that are used for power grid management tools, e.g., as with respect to power flow physics, uncertainty management or increased role of inverter-based resources).
3. Clearly differentiate between the appropriate uses of short-run and long-run marginal emissions rates in guidance documents as they are differently suitable for various applications. Specifically, how they *marginal* quantities relate to computing *total* emissions reductions.
4. Establish standardized protocols for data quality assurance that will be accepted for all parties and independent verification of marginal emissions calculations, including the ex-post evaluation of inaccuracies in estimation/prediction practices.

References

[1] W. W. Hogan, "Scope 2: Physical power usage accounting is fictional,pricing and marginal impact accounting are real," Working Paper / Report GHCP-121225, Harvard Kennedy School / Harvard Electricity Policy Group, 2025.

[2] Federal Energy Regulatory Commission, "Fact sheet — ferc directs nation's largest grid operator to create new rules to embrace innovation and protect consumers," Dec. 2025. Accessed: 2025-12-22.

[3] S. E. Sofia and Y. Dvorkin, "Carbon impact of intra-regional transmission congestion," *Cell Reports Sustainability*, 2025.

[1]Note that in radial networks, due to their simpler topology, it may be feasible but still face challenges, e.g., if a node hosts load, storage, distributed resources and thus operates as both a source and sink, that make it impractical.